

University of Florida Blitz Competition Team

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Introduction

- During the 2010s, the NFL experienced a passing boom, peaking in 2016 at an average of 241.5 passing yards per game.
- In recent years, that number has declined significantly, dropping to just 209.7 passing yards per game in 2025.
- At the same time, rushing production has shown a smaller but noticeable increase from the 2010s into the 2020s, signaling a subtle shift in offensive balance.

Introduction

Many analysts and fans attribute various recent offensive trends to the increased use of Cover 2 and other two-high safety defenses across the NFL. In this presentation, we'll examine how Cover 2 specifically impacts offensive production compared to other defensive schemes and explore what this evolution means for today's game.



Thesis

- Cover 2 at the NFL level comes down to situational run/pass rates.
- Cover 2 usage is most heavily influenced by down, distance and game script
 - Defenses are more likely to run cover 2 when they expect pass
- In general, teams can effectively beat Cover 2 by focusing on targeting the intermediate and deep portions of the middle of the field
- Defenses who employ 2 high safety looks are vulnerable to the run

Definitions

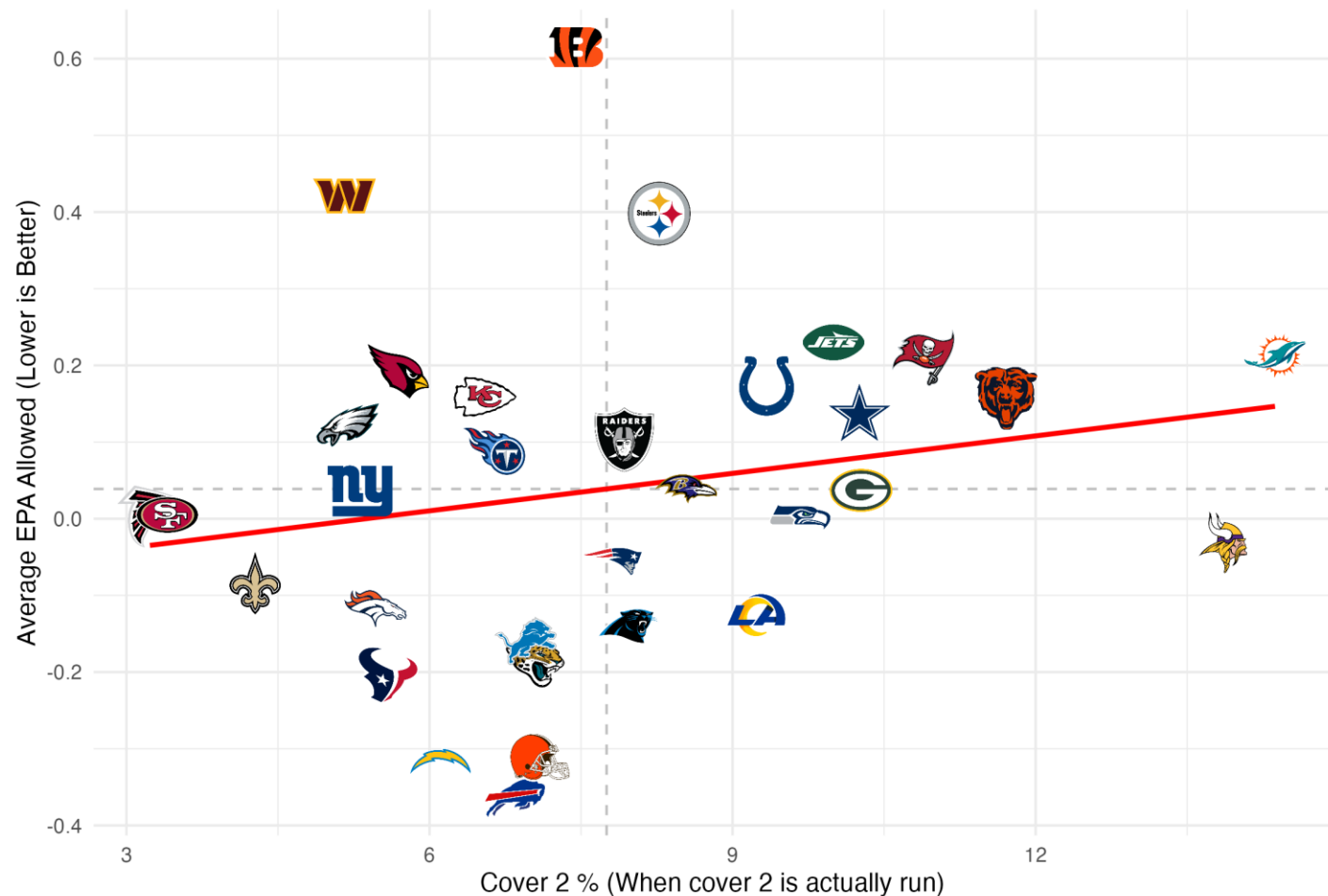
- Successful play
 - Gain of 40% of yards needed on 1st down, 60% of yards needed on 2nd down, and 100% of yards needed on 3rd and 4th down
- Cover 2
 - CoverageType tagged as Cover 2 or Man Cover 2
- Run/Pass situations are defined as down and distances where run/pass rate is >65%.
 - In 2025, run situations were 1st & 1-5, 2nd & 1-3, and 3rd & 1-2.
 - Pass situations were 1st & 11+, 2nd & 8+, and 3rd and 3+.
 - With this in mind, we can look at how Cover 2 usage changes.

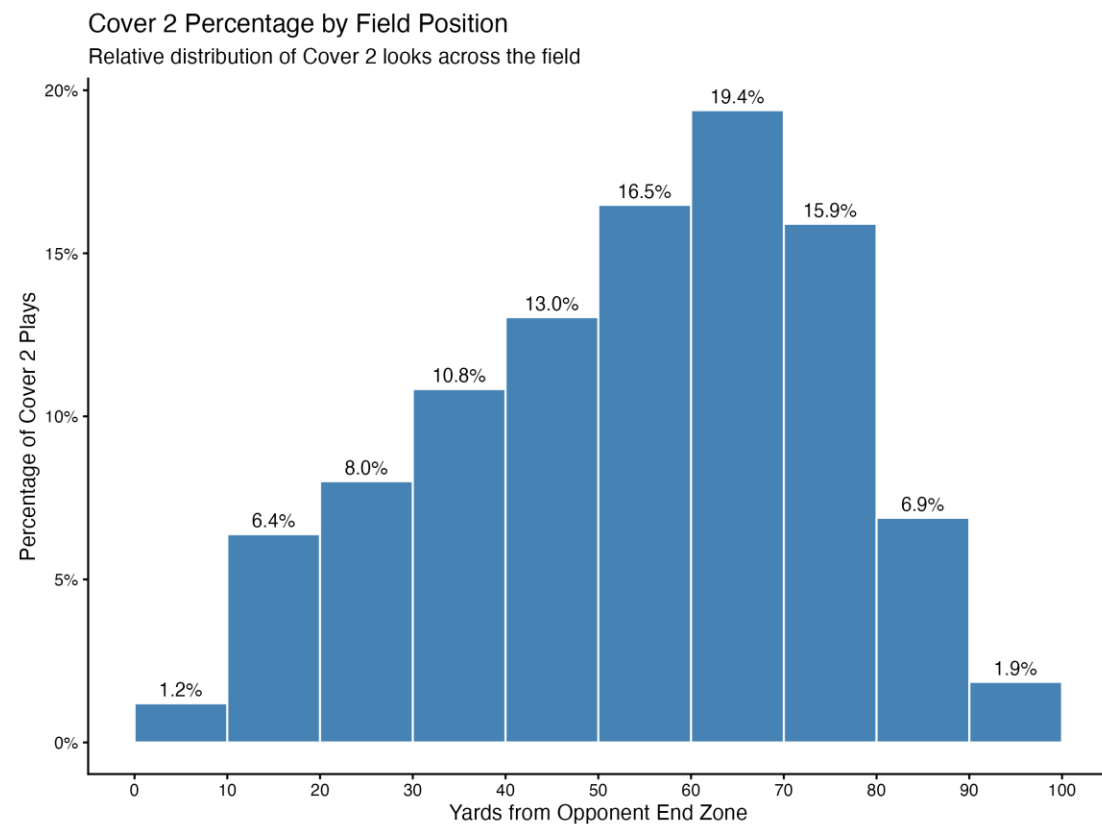
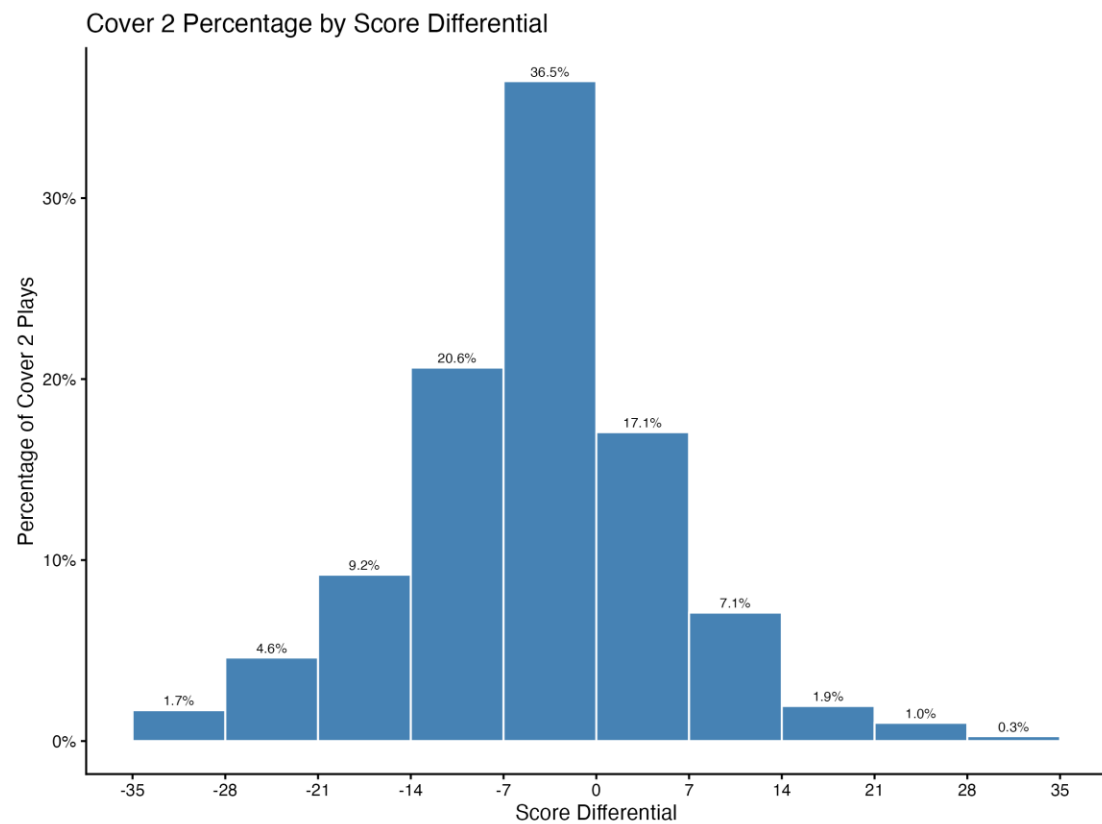


Part 1

Understanding Two-High Safety Coverage Deployment and League Trends

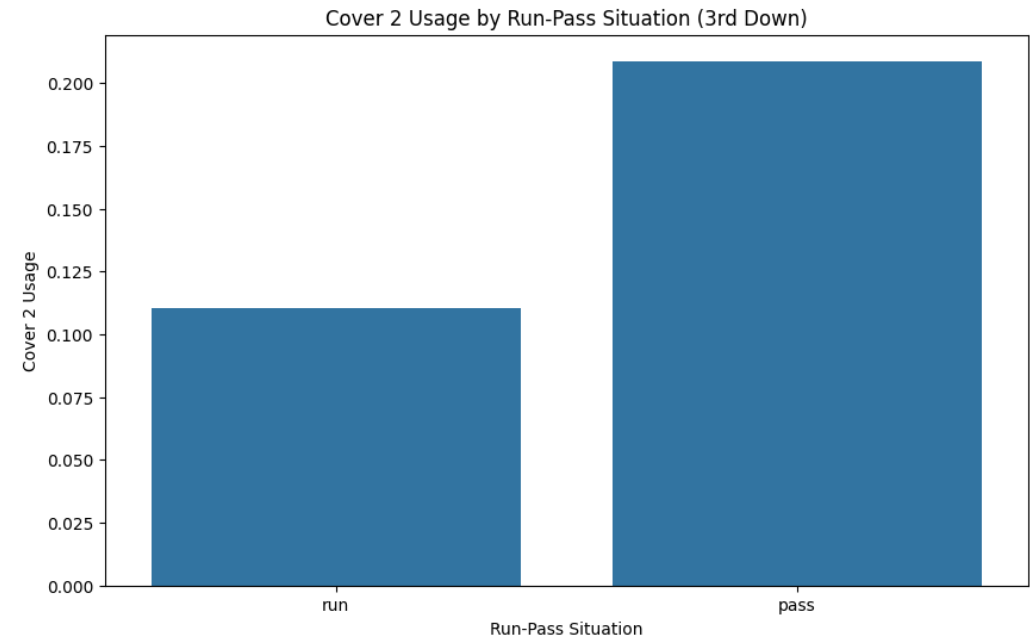
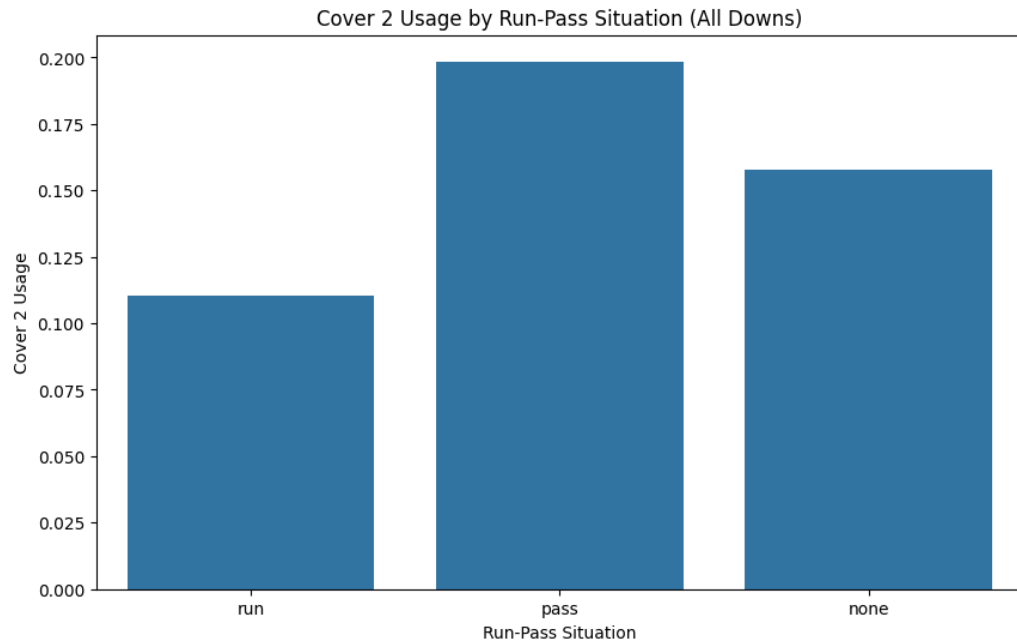
Dashed lines are league averages





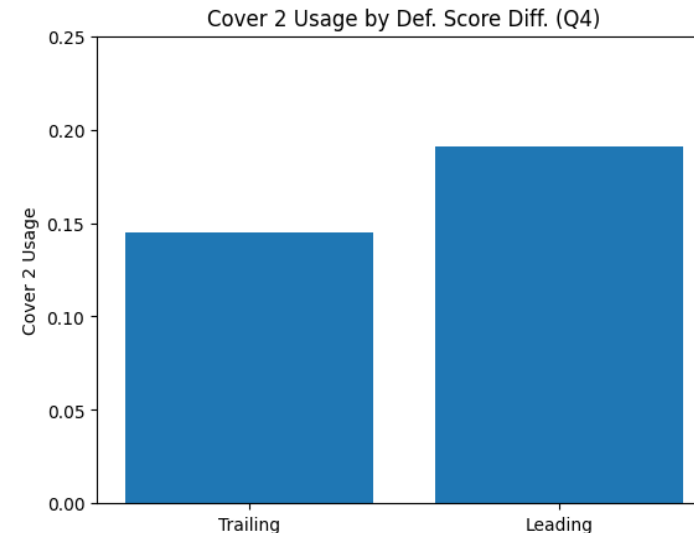
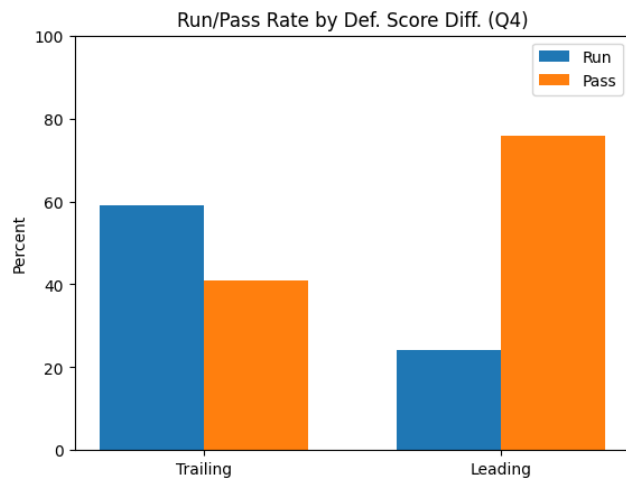
Cover 2 Usage by Situation

- Observe how Cover 2 usage increases in passing situations
 - Even larger increase on 3rd down



Score Differential

- Grouping by quarter, Cover 2 usage is relatively uniform
- When considering score differential, we see a 5% higher Cover 2 rate when the defense is leading
 - Run rate when trailing is 12% lower compared to when leading
- Quarter 4 reflects this trend



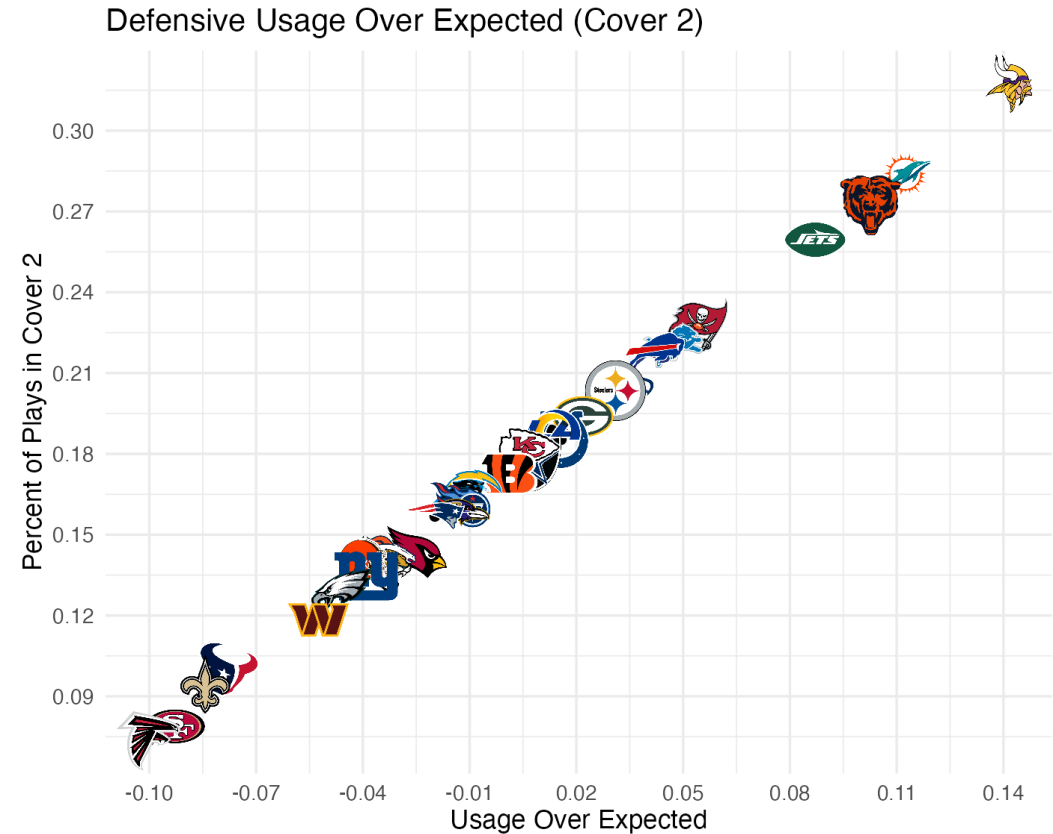
Usage Over Expected

- To see which teams use Cover 2 the most, it would be best to see who uses it more than in unusual situations
 - Shows teams who use Cover 2 because they want to, not because it's the consensus choice
- Grouped by situation (down, distance, field position), find average Cover 2 usage in by situation
- Difference between actual usage (1 or 0) and average usage constituted above/below average
 - Based on SumerSports article [1]

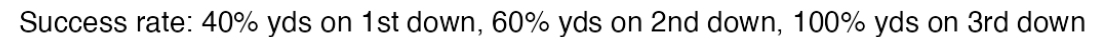
[1] <https://sumersports.com/the-zone/sumersports-in-game-coaching-model-an-introduction/>

Defensive Cover 2 Usage

- Teams use general Cover 2 based on overall team tendency
- Teams stick to generally uniform situational usage



Red lines are league averages
Correlation: 0.37



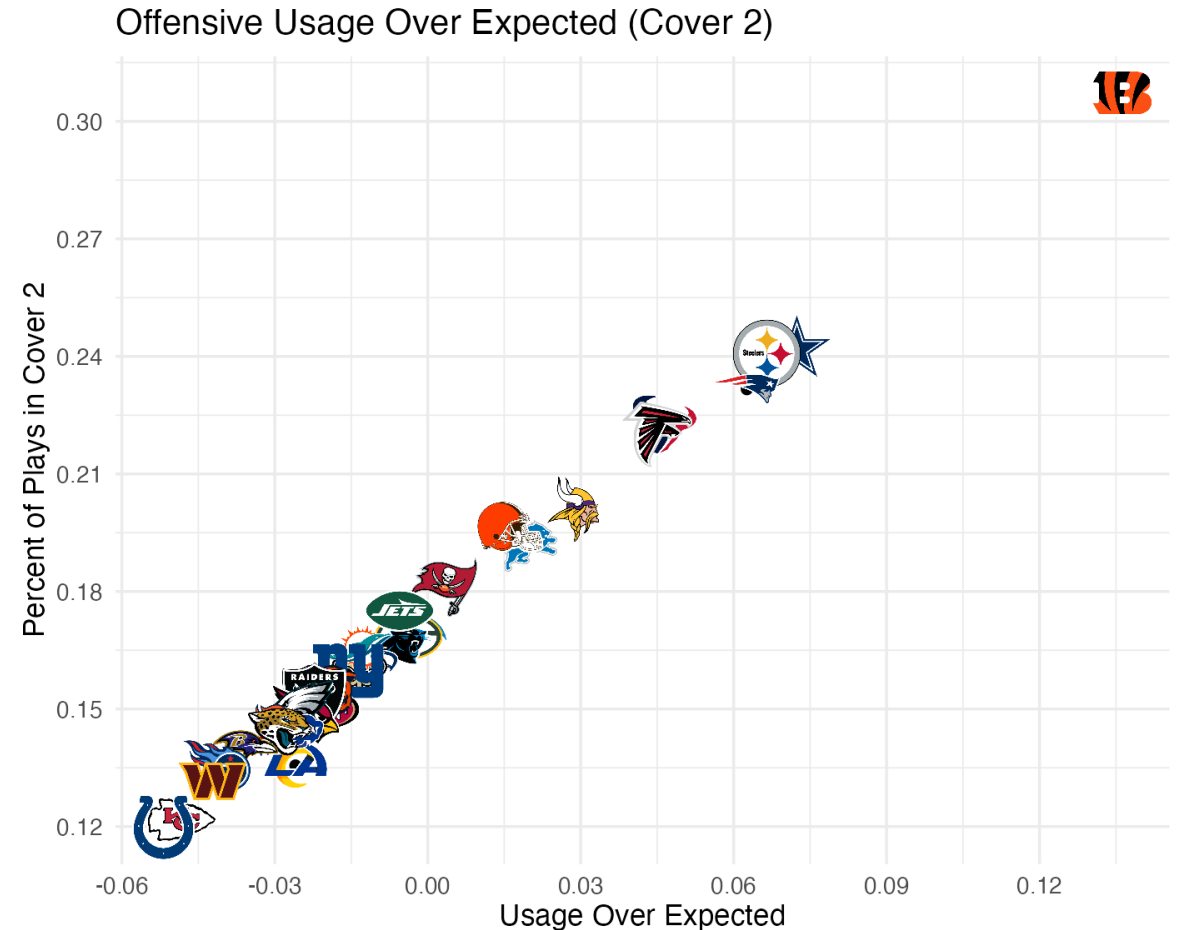
Part 2

Offensive Attack Strategies Against Two-High Coverage



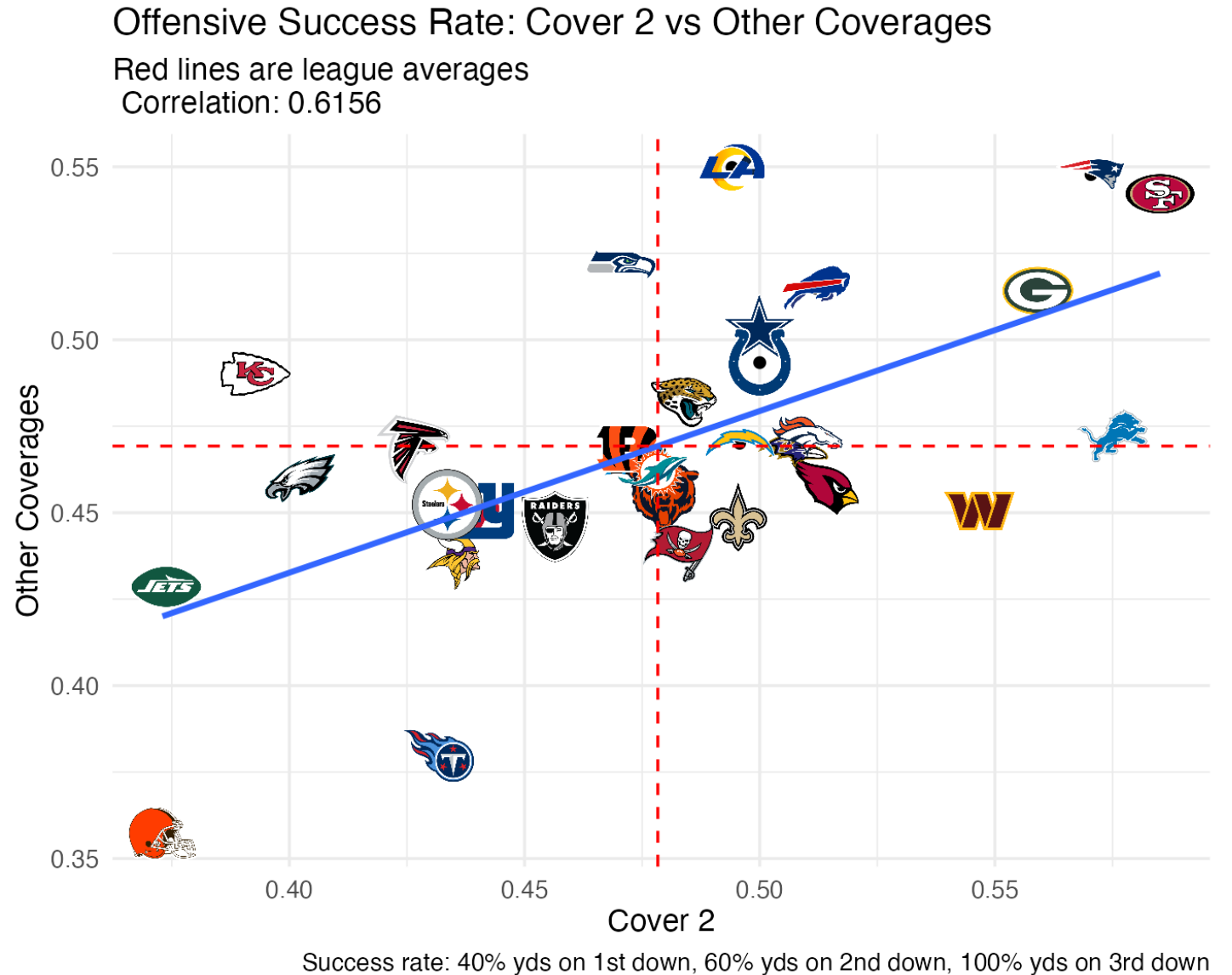
Offensive Cover 2 Faced

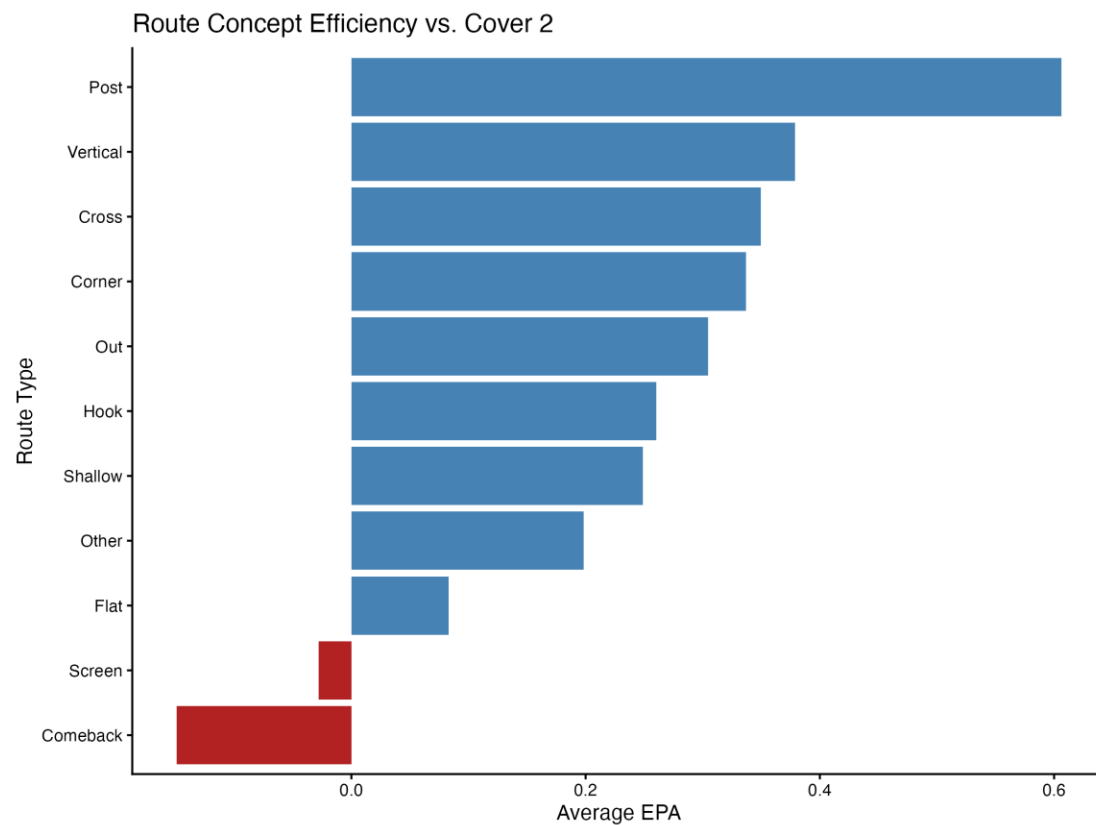
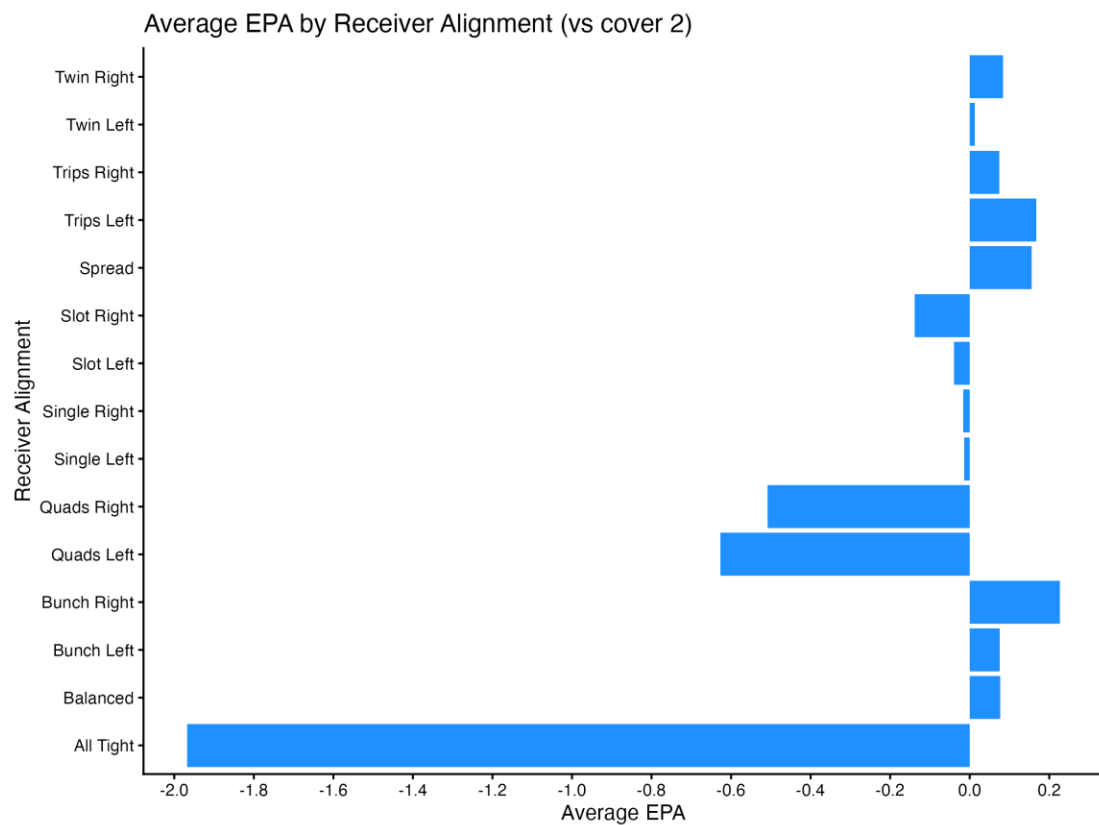
- Like defensive usage, teams face Cover 2 based on their offensive profile, with little variation based on situation



Offensive Effectiveness

- NE, SF, GB are generally good offenses, independent of coverage type
- KC, SEA struggled with specifically with Cover 2
- WAS, ARI excel against Cover 2, but lacked otherwise
- CLE, TEN, NYJ struggled against all coverages
- General positive linear trend with consistent team success





But Why is the Run the Enemy of Cover 2?

- While coverage data isn't provided on true run plays, we can use # of safeties to estimate the potential use of Cover 2
 - 75% of Cover 2 plays start with 2 high safeties
- When breaking down run plays by # of safeties, a trend emerges
 - Preparing to use Cover 2 (starting with 2 high safeties) leaves you vulnerable to the run.

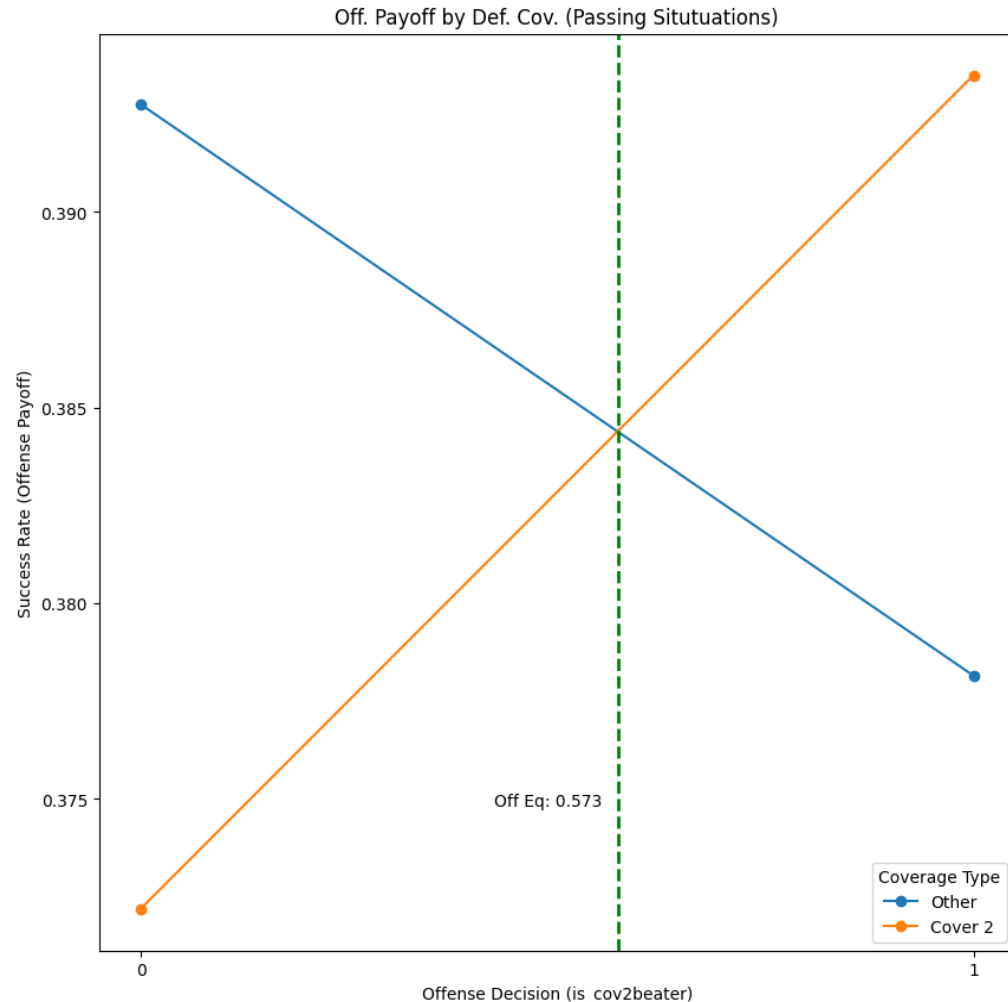
# Safeties	Count	Success %	EPA/Play
0	1264	47.9%	-0.117
1	6413	47.3%	-0.056
2	5572	50.3%	-0.013
3	227	46.7%	-0.056

Varying your Pass Concept

- In situations where you'd expect to see Cover 2, it may seem optimal to consistently run a concept that beats Cover 2
 - What about when the defense catches on and neutralizes your concept?
- Game theory has a solution: Nash Equilibrium - “a situation where no player could gain more by changing their own strategy (holding all other players' strategies fixed) in a game” [2].
- Because Cover 2 is most used in passing situations, we will focus there

[2] https://en.wikipedia.org/wiki/Nash_equilibrium

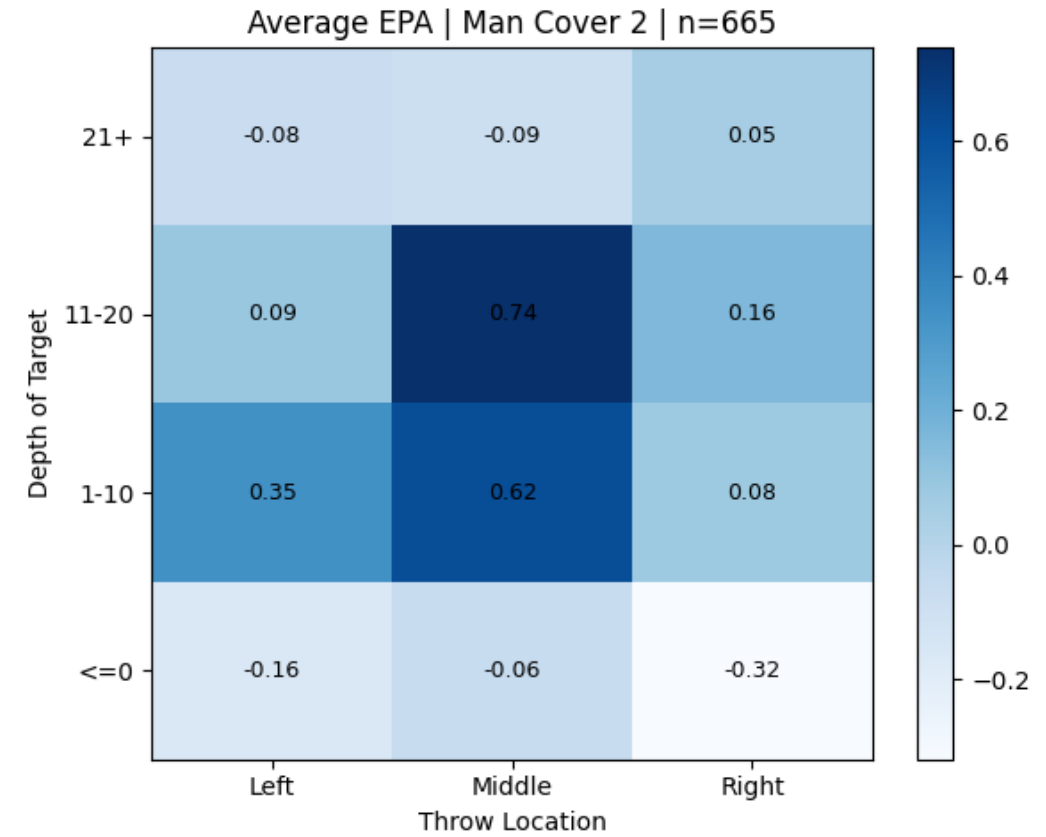
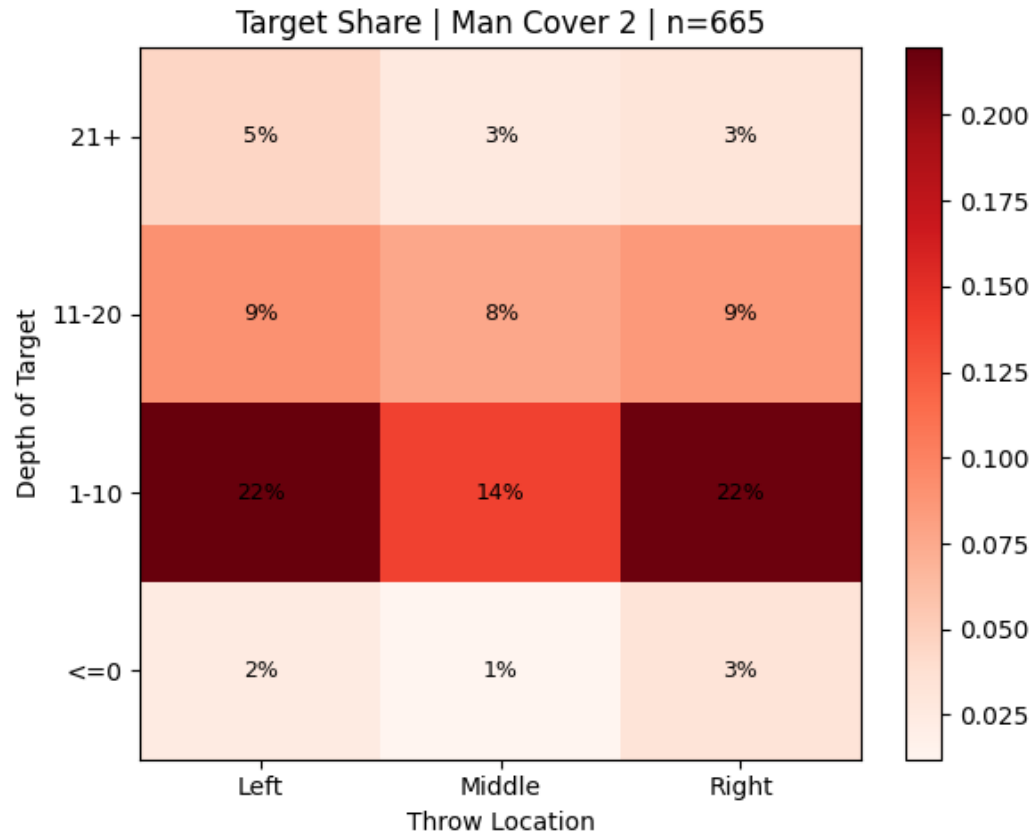
Optimal Offensive Usage in Passing Situations



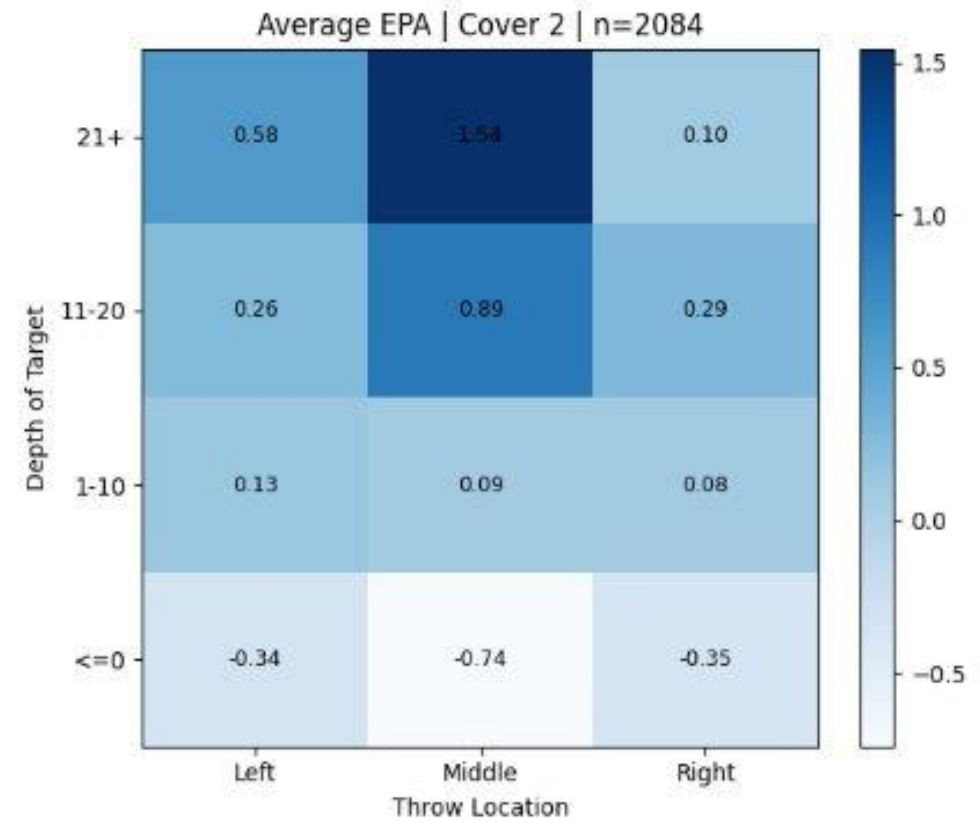
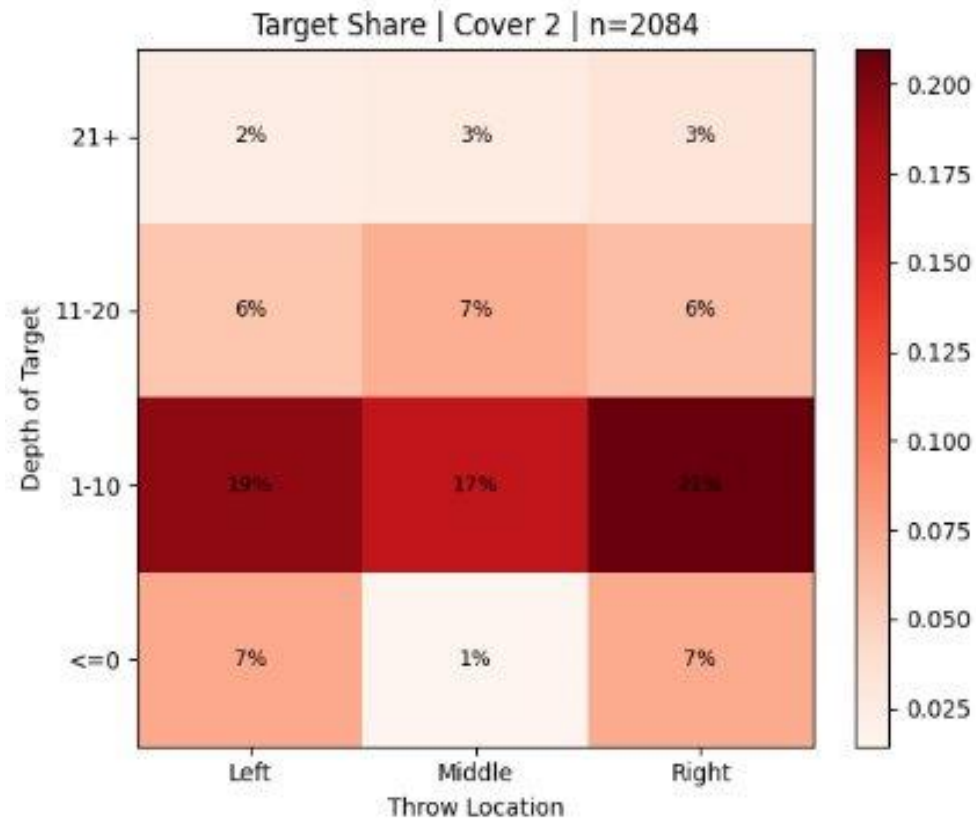
- On pass plays, offense should run Cover 2 beater 57% of the time
 - Maximizes EPA/Play when assuming defense acts rationally (is making optimal choices)

Cover 2 beaters defined as plays targeting post, vertical, and crossing routes

Cover 2 Man Heat Map



Cover 2 Zone Heat Map



PART 3

Situational Game Planning and
Specific Play Design



Team Comparison – Cover 2

- In general, Minnesota is less successful when running Cover 2 compared to standard coverage
 - Tied to situational factors and heavy usage rate
- NE is strong across both coverage categories
 - Bolstered by strong deep attack

NE Offense		
Coverage	Success Rate	EPA
Cover 2	0.570423	0.319339
Other	0.547771	0.2537

MIN Defense		
Coverage	Success Rate	EPA
Cover 2	0.467066	0.038214
Other	0.406685	-0.190917

Team Comparison - Personnel

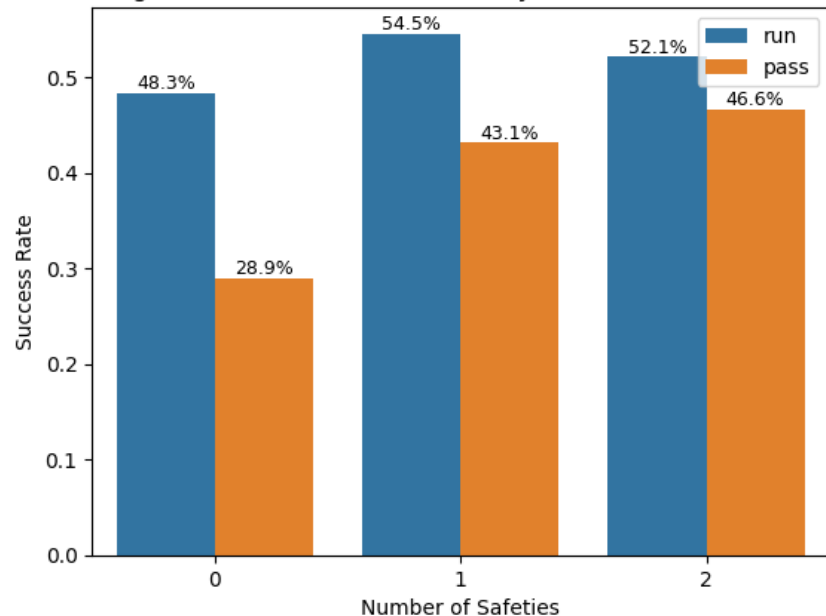
- NE is an elite 11 personnel team, while MIN is elite against 11 personnel
 - Aligns with how Cover 2 is good against 11 personnel
 - When NE gets into 11 personnel, MIN will match with either **Base** or Nickel
- **NE wants to play MIN in Base**
 - NE is good against Base (45% success rate)
 - MIN is bad in Base(52% success rate allowed)
 - MIN matches **12 and 13 personnel** with base 79% of the time

Defensive Box Player Usage (MIN)	
Box Players	Percent of Plays
0	0.098
4	0.195
5	8.000
6	35.024
7	52.878
8	3.512
9	0.293

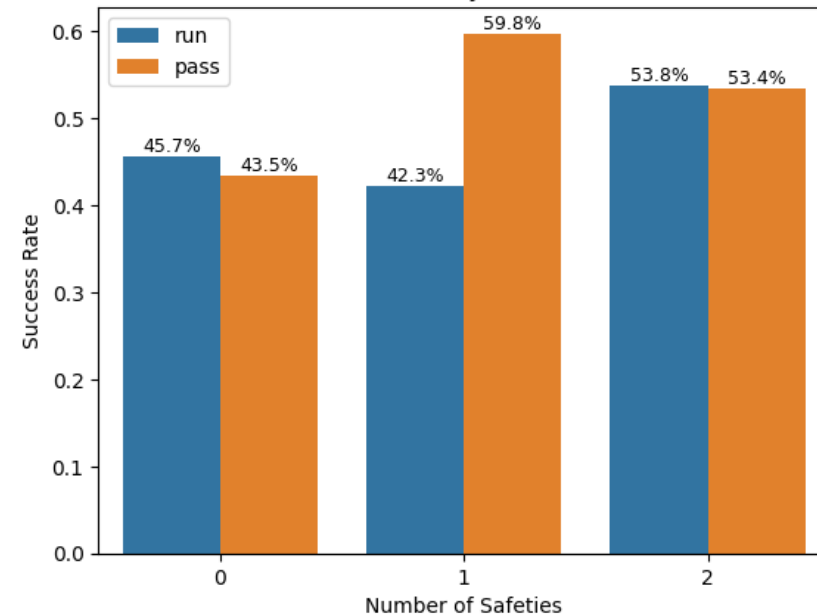
Team Comparison – Safeties

- MIN runs 2 high safeties pre-snap 53% of the time, with 71% of their Cover 2 snaps coming from this look.
 - NE should run the ball when they see a 2 high look to exploit mismatch

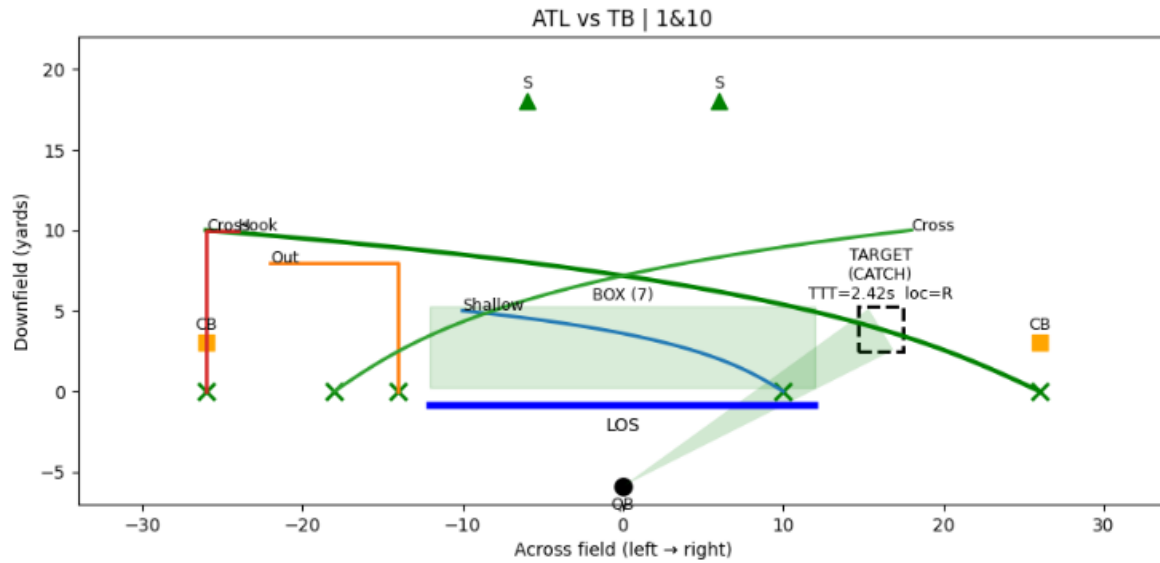
Vikings Def. Success Rate Allowed by # of Safeties (All Downs)



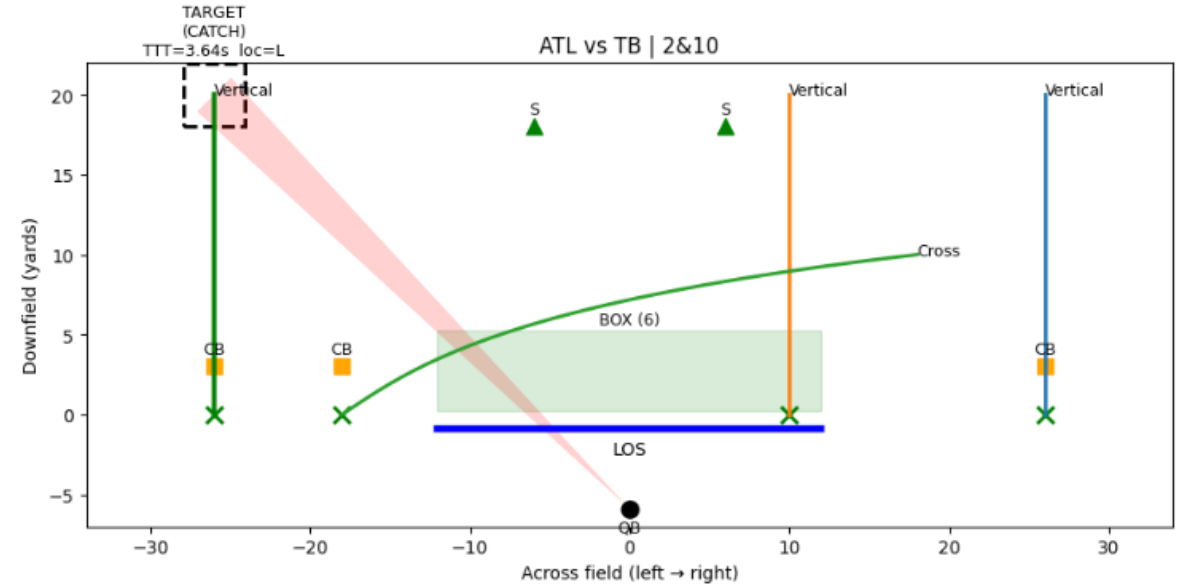
Patriots Off. Success Rate by # of Safeties (All Downs)



Play Visualization



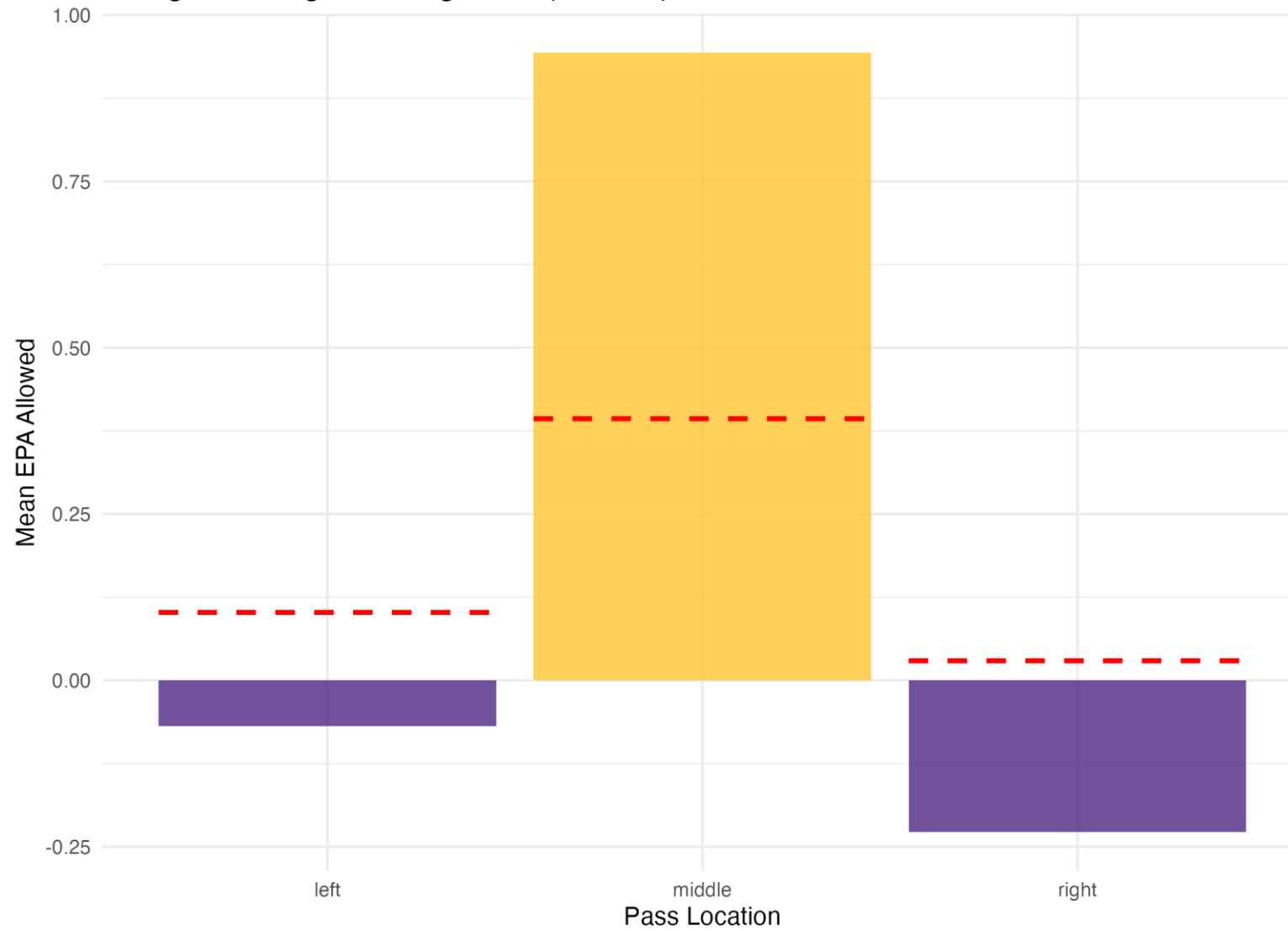
https://www.youtube.com/watch?v=A_EbbsPORqg (0:00)



https://www.youtube.com/watch?v=A_EbbsPORqg (13:48)

Offensive and Defensive alignment from personnel and R1-L1, RWR-LWR columns

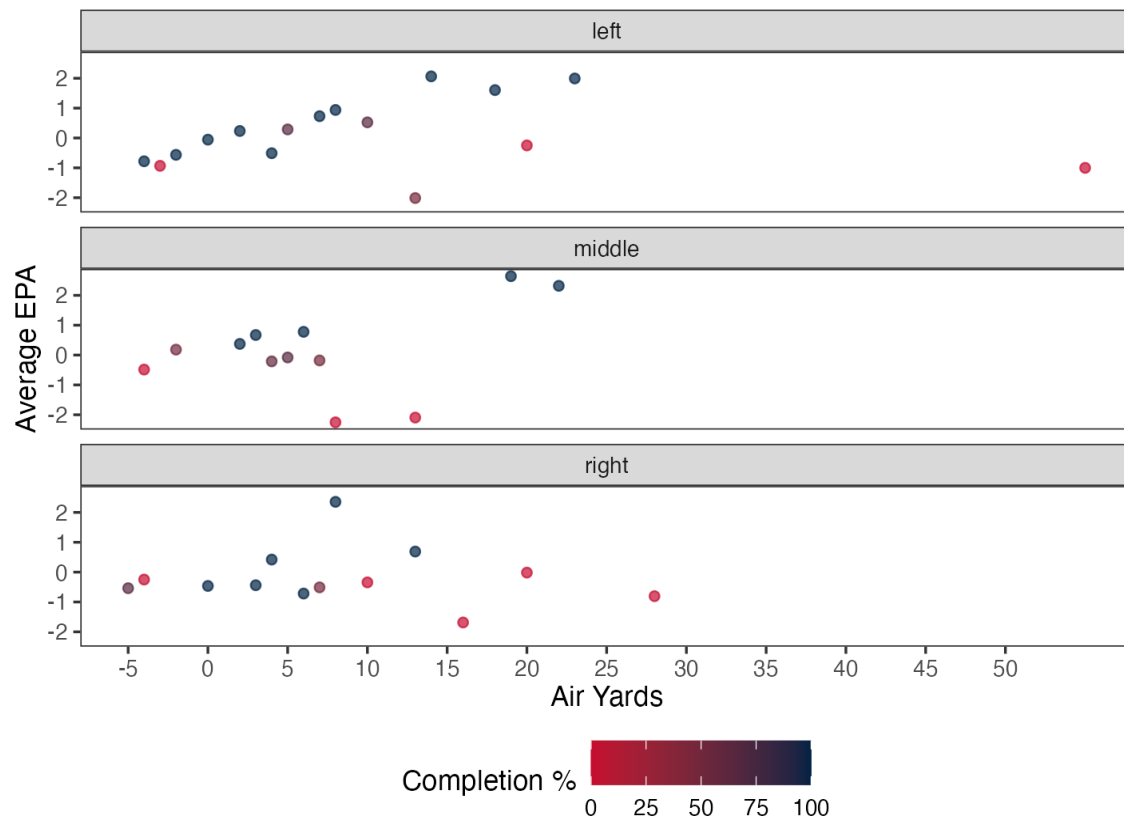
Vikings vs. League Average EPA (Cover 2)





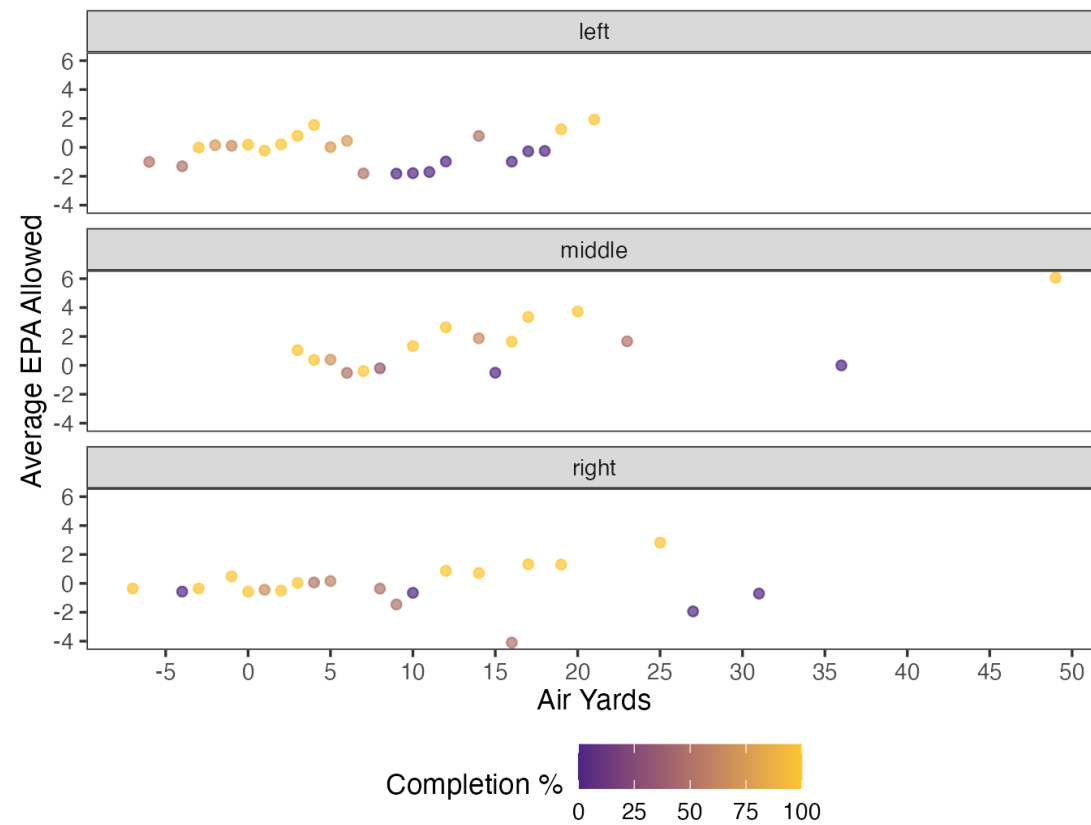
Offensive Efficiency by Air Yards (NE)

When Playing Against Cover 2 / Cover 2 Man



Defensive Efficiency by Air Yards (MIN)

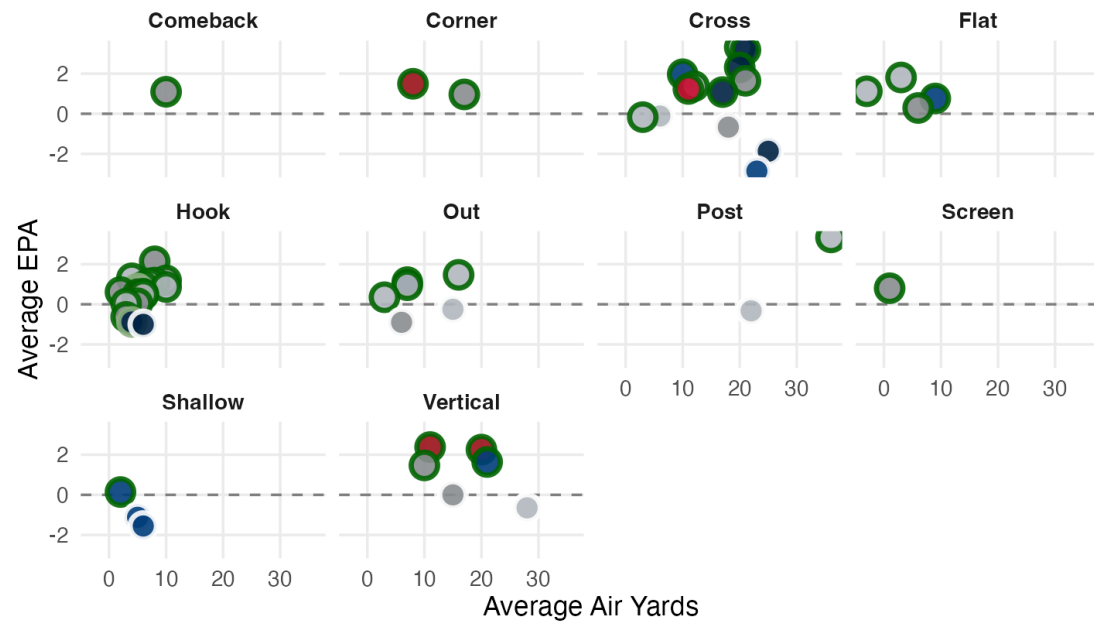
When Playing Cover 2 / Cover 2 Man





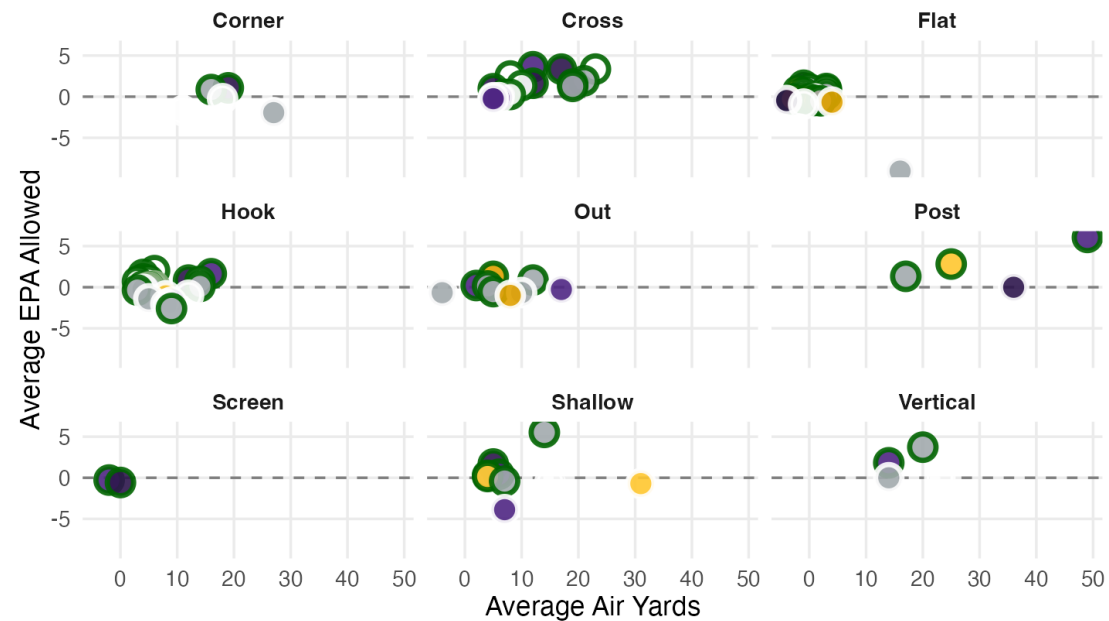
NE Route Efficiency by Position

Targeted Players, when facing Cover 2 / Cover 2 Man



MIN Route Efficiency Allowed by Position

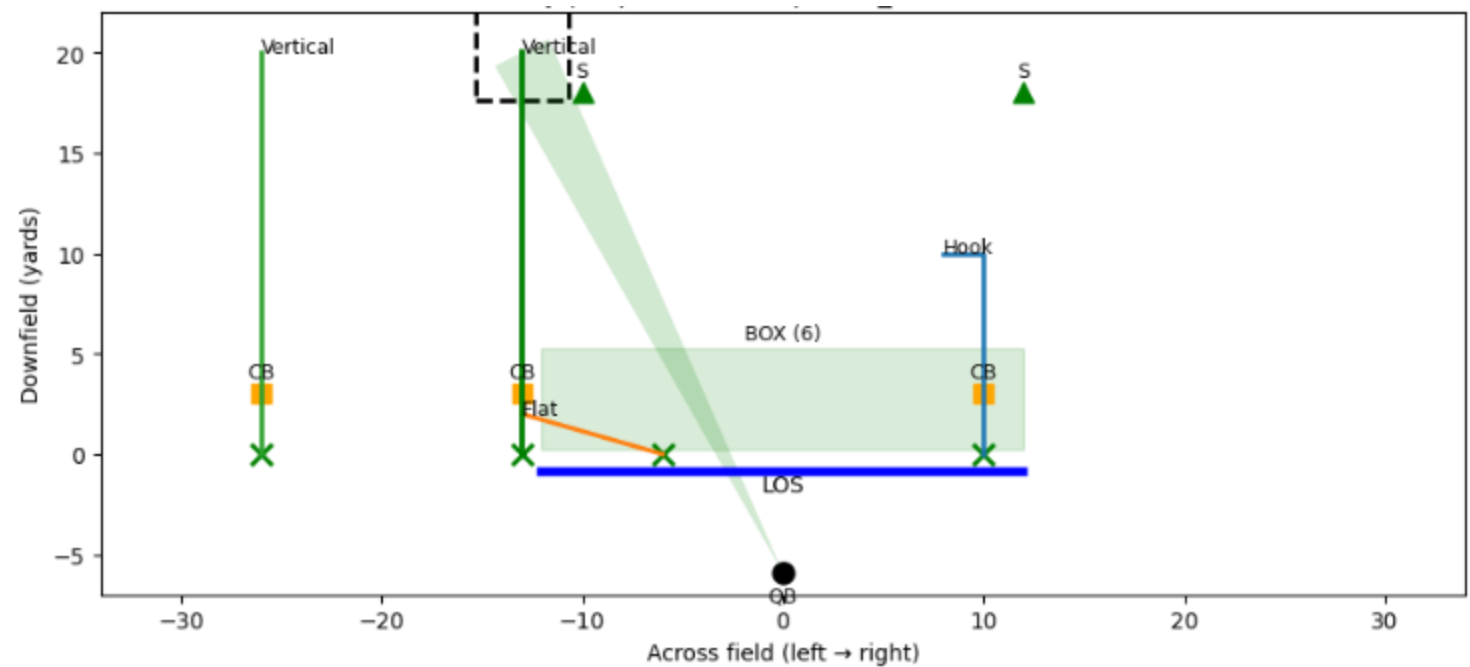
Targeted Players, when in Cover 2 / Cover 2 Man



Offensive Gameplan Summary

- Get into heavy personnel
 - MIN is not comfortable matching up with 12 and 13 personnel packages
- Run when you see two high safety looks
 - NE matches up well with running against these defensive alignments
- Minnesota gives up a lot of big plays over the middle (way above the league average)
 - Patriots aren't great at throwing the deep ball, but they complete a lot of passes. Their outside receivers do not get targeted on deep routes
- Crossers and Post Routes are most effective (NE is great at them, MIN gives them up)
 - Screen game is not effective against Cover 2

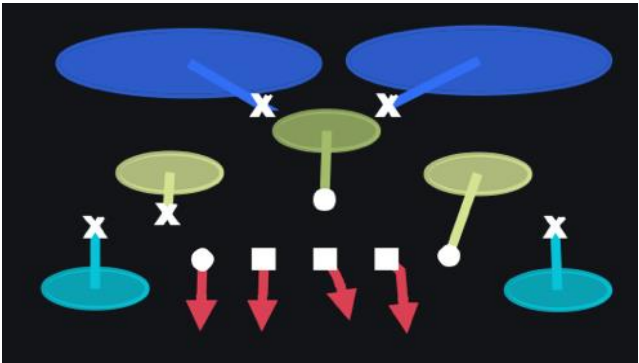
Optimal Pass Play Against Cover 2 Zone



Optimal targeted route



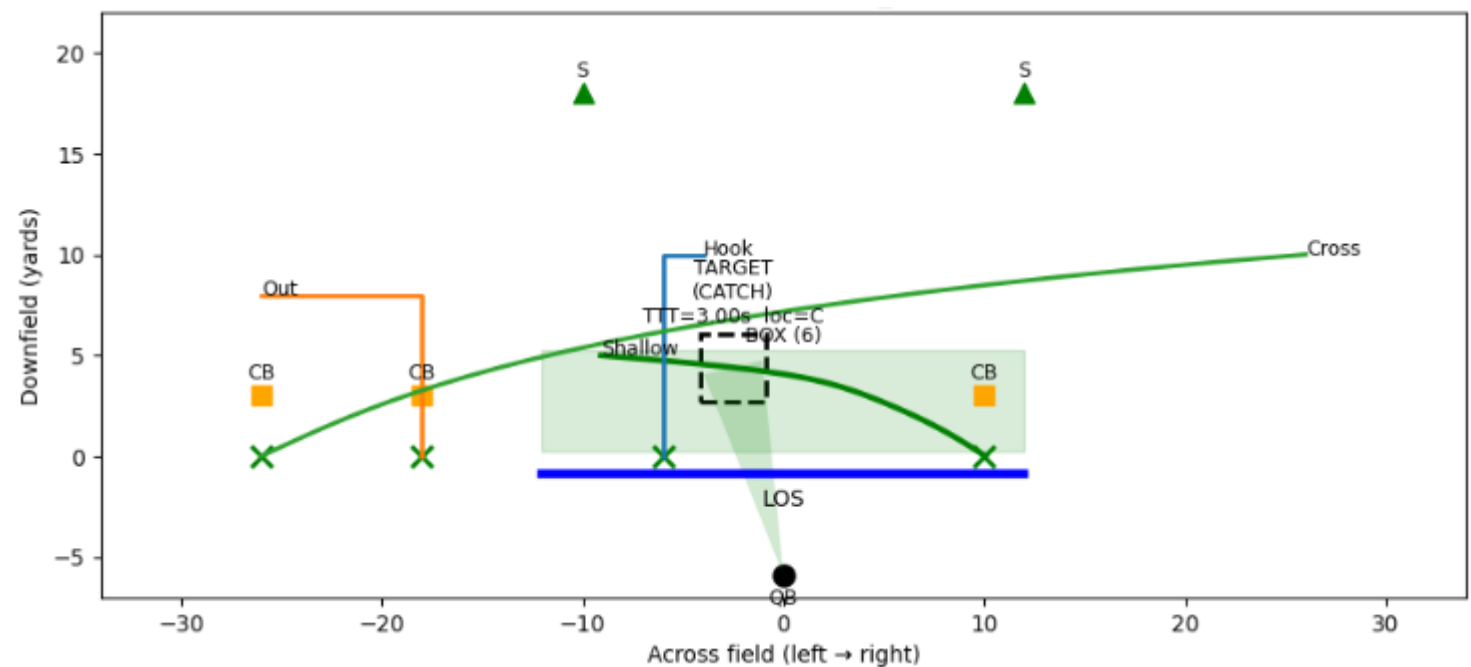
Location	Route	Number of Plays	Mean EPA	Median EPA
L1	Cross	84	0.679954	1.093516
L2	Vertical	40	0.882858	0.90827
L3	Hook	72	0.422929	0.32062
R1	Corner	26	0.681712	0.998018
R2	Vertical	38	0.881236	1.079111
R3	Hook	67	0.117446	0.21482



Routes ran on plays with L2 Vertical

Location	Route	Plays	Mean EPA	Media EPA
L1	Vertical	15	1.659863	1.2581
L1	Post	4	1.212073	1.748301
L1	Hook	7	0.675917	1.231441
L3	Flat	4	3.249805	3.082934
L3	Hook	3	-0.131197	-0.248794
L3	Cross	3	-0.184403	-0.035645
R1	Hook	6	1.483921	1.241214
R1	Vertical	20	1.094911	1.170819
R1	Shallow	4	0.682336	0.611228
R2	Hook	3	1.222751	1.573054
R2	Vertical	13	1.175191	1.2581

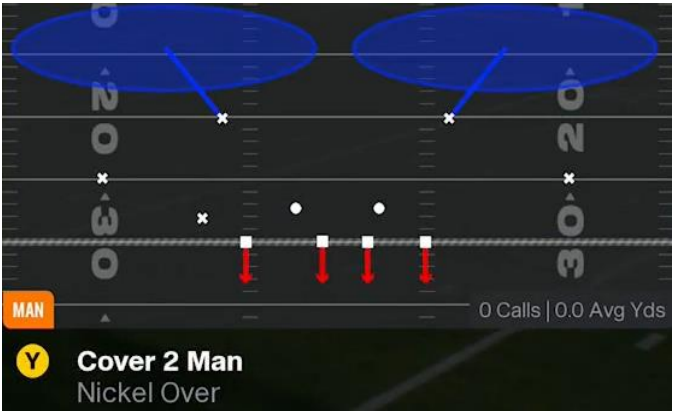
Optimal Pass Play Against Cover 2 Man *Much Lower Sample



Optimal targeted route



Rank	Route	Plays	Mean EPA	Median EPA
L1	Hook	22	0.531082	0.474862
L2	Out	31	0.313328	0.330839
R1	Shallow	46	0.524855	0.275762
R2	Out	24	0.402962	0.066194



Routes ran on plays with R1 Shallow

Location	Route	Plays	Mean EPA	Media EPA
L1	Cross	8	1.337292	1.922525
L1	Post	4	1.078498	1.327002
L1	Vertical	12	0.625109	0.857137
L2	Out	9	0.984253	1.391645
L2	Hook	5	0.952157	1.9943
L2	Cross	6	0.935189	0.813878
L3	Hook	3	1.491611	2.248789
L3	Vertical	4	1.389748	1.889318
R2	Vertical	3	0.285433	0.467076
R2	Out	3	0.2571	0.200924